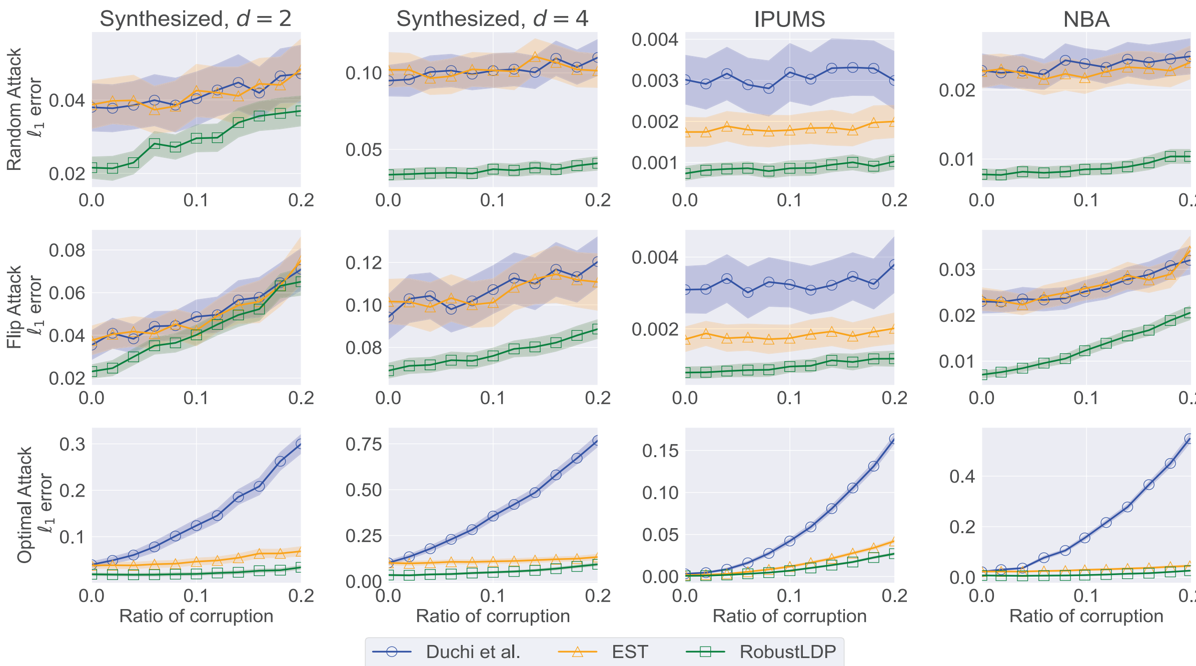

Figure 9: Comparison of ℓ_2 mean estimation methods. We set $\epsilon = 10$ for all experiments.

We use a synthesized dataset with $d = 8$. The results are shown in Figure 10.

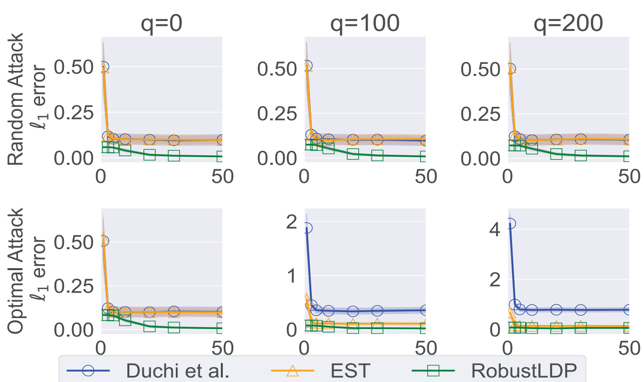

Figure 10: The ℓ_2 error of mean estimation over ℓ_2 support with respect to ϵ.

From Figure 10, it can be observed that the original method in [2] does not perform well, especially with manipulation. Compared with the original method, the EST method in [1] exhibits significantly lower estimation error. However, Figure 10 also shows that with the increase of ϵ, the error of EST does not converge to zero, even if there are no attacks. Compared with EST, our new method achieves nearly the same performance as EST for small ϵ. With larger ϵ, our method has significantly smaller ℓ_2 error than EST, indicating that our method has advantages in large ϵ. The results agree with our theoretical analysis.

9. Related Work

LDP protocols without attacks. For frequency estimation, [3] proposed Rappor. [15] proposed OUE and OLH. [17] and [18] proposed optimal LDP protocols at all privacy regimes and has improved communication complexity. For mean estimation, [2] and [19] proposed estimators that project each sample on the surface of a sphere. Improved methods are then proposed in [20], [22]. [33] proposed a refined mean estimator under LDP. Apart from these basic problems, LDP has been used in some advanced tasks [39], [40], [41], [42], [43].

Manipulation attacks against LDP protocols. LDP protocols are well known to be vulnerable to poisoning attacks [1], [10], [44]. [10] proposed three common attacks: Random Perturbation Attack (RPA), Random Item Attack (RIA) and Maximal Gain Attack (MGA). [44] extended these attacks to key-value data. [1] designed an optimal untargeted attack strategy. [45] proposed an attack that can distort both mean and variance estimation.

Defense against manipulation attacks. Cao et al. [10] proposed several standard methods. [9] proposed LDPGuard. [11] proposed LDPRecover, which attempts to recover the real frequency from its corrupted estimation. [1] proposed general defense strategies, called HST and EST, for frequency estimation and ℓ_2 mean estimation, respectively. [46] generalizes the analysis to general information constraints.

10. Conclusion

In this paper, we have designed a new robust estimation framework under LDP that can withstand output poisoning attacks. We discuss three common tasks: frequency estimation, ℓ_1 and ℓ_2 mean estimation. For frequency estimation, each element of the pre-defined signal can take k possible values. For ℓ_1 and ℓ_2 mean estimation, there are k independent signals for each user. Our theoretical analysis